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Flashlight with an electronic insect control system

Abstract

Exemplary embodiments of a flashlight having an electronic insect control system are provided. In some exemplary embodiments, a flashlight apparatus is provided, having a head portion comprising a first light source, a tail portion, and a body portion between the head portion and tail portion, the body portion comprising an electronic insect control system, the electronic insect control system comprising a frame portion, a second light source provided within the frame portion, and an electrical grid provided within the frame portion and surrounding the second light source, wherein the electrical grid is configured to generate a voltage.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of and claims benefit to U.S. patent application Ser. No. 18/138,291, filed on Apr. 24, 2023, entitled “Flashlight with an Electronic Insect Control System”, which is a continuation of and claims benefit to U.S. patent application Ser. No. 17/513,825, filed on Oct. 28, 2021, entitled “Flashlight with an Electronic Insect Control System”, which is a continuation of and claims benefit to U.S. patent application Ser. No. 16/900,512, filed on Jun. 12, 2020, entitled “Flashlight with an Electronic Insect Control System”, which relates to and claims priority from U.S. Provisional Patent Application Ser. No. 62/939,435 filed Nov. 22, 2019, and United States Design application Ser. No. 29/694,652 filed Jun. 12, 2019, the entire disclosures of which are hereby incorporated herein by reference.

FIELD OF THE DISCLOSURE

(1) The present disclosure relates to exemplary embodiments of a flashlight, and more particularly, to exemplary embodiments of a flashlight having an electronic insect control system.

BACKGROUND INFORMATION

(2) Various flashlights can be provided for outdoor use. When using flashlights in the dark, insects are often attracted to the light which can be bothersome to the user holding the flashlight. The present disclosure relates to a flashlight that can be used as a flashlight while providing an electronic insect control system.

SUMMARY OF EXEMPLARY EMBODIMENTS OF THE DISCLOSURE

(3) Exemplary embodiments of the present disclosure can provide for a flashlight that can have an electronic insect control system, having a light portion at a first end, and an additional light source that provides an electronic insect control system.

(4) For example, in some exemplary embodiments, a flashlight apparatus is provided, comprising a head portion comprising a first light source, a tail portion, and a body portion between the head portion and tail portion, the body portion comprising an electronic insect control system, the electronic insect control system comprising a frame portion, a second light source provided within the frame portion, and an electrical grid provided within the frame portion and surrounding the second light source, wherein the electrical grid is configured to generate a voltage.

(5) In some exemplary embodiments, the flashlight apparatus further comprises one or more batteries configured to power the first light source, second light source and electrical grid. The flashlight apparatus can further comprise a USB port for charging the one or more batteries. In some exemplary embodiments, the first light source comprises a printed circuit board including one or more light emitting diodes, and the second light source comprises a fluorescent lamp. In some exemplary embodiments, the frame portion comprises a plurality of holes providing access to the electrical grid and second light source. In some exemplary embodiments, the electrical grid comprises one or more wires that surround the second light source.

(6) In some exemplary embodiments, the flashlight apparatus further comprises a cover on the body portion, the cover configured to enclose the electronic insect control system. The cover can be a sliding cover that encloses the electronic insect control system in a first position and reveals the electronic insect control system in a second position. In some exemplary embodiments, the flashlight apparatus further comprises a slider provided on the sliding cover configured for a user's finger to slide the sliding cover from the first position to the second position.

(7) In some exemplary embodiments, a flashlight apparatus is provided that comprises a head portion comprising a first light source, a tail portion, a body portion between the head portion and tail portion, the body portion comprising an electronic insect control system, and a sliding cover on the body portion, the sliding cover configured to enclose the electronic insect control system in a first position and reveal the electronic insect control system in a second position. In some exemplary embodiments, the flashlight apparatus further comprises a slider provided on the sliding cover configured for a user's finger to slide the sliding cover from the first position to the second position. In some exemplary embodiments, the electronic insect control system comprises a frame portion, a second light source provided within the frame portion, and an electrical grid provided within the frame portion and surrounding the second light source, wherein the electrical grid is configured to generate a voltage.

(8) In some exemplary embodiments, the flashlight apparatus further comprises one or more batteries configured to power the first light source, second light source and electrical grid. In some exemplary embodiments, the flashlight apparatus further comprises a USB port for charging the one or more batteries. In some exemplary embodiments, the first light source comprises a printed circuit board including one or more light emitting diodes, and the second light source comprises a fluorescent lamp.

(9) In some exemplary embodiments, the frame portion comprises a plurality of holes providing access to the electrical grid and second light source. In some exemplary embodiments, the electrical grid comprises one or more wires provided in a spiral configuration that surround the second light source. In some exemplary embodiments, the body portion is configured to be held in a user's hand.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The foregoing and other objects of the present disclosure will be apparent upon consideration of the following detailed description, taken in conjunction with the accompanying drawings, and claims, in which like reference characters refer to like parts throughout, and in which:

(2) FIG. 1 is a perspective view of a flashlight having an electronic insect control system according to an exemplary embodiment of the present disclosure;

(3) FIG. 2 is a side view of a flashlight having an electronic insect control system according to an exemplary embodiment of the present disclosure;

(4) FIG. 3 is a side view of a flashlight having an electronic insect control system in an uncovered configuration according to an exemplary embodiment of the present disclosure; and

(5) FIG. 4 is a perspective view of a flashlight having an electronic insect control system in a covered configuration according to an exemplary embodiment of the present disclosure.

(6) Throughout the figures, the same reference numerals and characters, unless otherwise stated, are used to denote like features, elements, components or portions of the illustrated embodiments. Moreover, while the subject disclosure will now be described in detail with reference to the figures, it is done so in connection with the illustrative embodiments. It is intended that changes and modifications can be made to the described embodiments without departing from the true scope and spirit of the subject disclosure.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS OF DISCLOSURE

(7) The exemplary embodiments of the present disclosure can provide for a flashlight that can have an electronic insect control system. The flashlight can have a lighting portion at a first end, and an additional light source that attracts insects to an electrical grid ("bug zapper"). The flashlight can be used with the bug zapper portion on or off independently of the light portion at the first end, or solely as a bug zapper. The bug zapper portion can be optionally covered.

(8) Exemplary embodiments of the various methods and apparatuses will now be described with reference to the figures. The following description of the various embodiments is merely exemplary in nature and is in no way intended to limit the scope of the disclosure, its application, or uses.

(9) FIG. 1 is a perspective view of a flashlight **100** and FIG. 2 is a side view of a flashlight **100** having an electronic insect control system **150** according to the exemplary embodiments of the present disclosure. The flashlight **100** can have a head portion **110**, a body portion **120** and a tail portion **130**. The head portion **110** can include a light **106** on an upper portion thereof. The light **106** can include a lens cover and a light source within for providing a first light at a first end **102** of the flashlight **100**. The light source can be one or more light bulbs or any type of light source, such as but not limited to a printed circuit board containing one or more light-emitting diodes. The light source can be provided within the head portion **110**. The head portion **110** can be threadably engaged with the body portion **120**. The tail portion **130** can be threadably engaged to the body portion **120** at a second end **104**, and can include a switch (not shown) at the second end **104**, which can be used to turn the light **106** on and off, as well as turn the electronic insect control system on and off, which will be further described below.

(10) The body portion **120** can include an upper body portion **122** and a lower body portion **124**. The lower body portion **124** can house one or more batteries for powering the light **106** and the electronic insect control system. The tail portion **130** can be threadably removed from the body

portion **120** to insert one or more batteries (e.g., alkaline battery) into the lower body portion **124**. In some exemplary embodiments, the batteries within the lower body portion **124** can be rechargeable (e.g., a lithium-ion battery), such that the tail portion **130** may not need to be threadably removable from the body portion **120** to change the one or more batteries.

(11) The body portion **122** can include a frame portion **126** at the upper body portion **122**, and a sliding cover **128**, shown at the lower body portion **124** in FIGS. **1** and **2**. The frame portion **126** can comprise a circular frame and can include an electronic insect control system **150** (“bug zapper”), which can be located internal of and in a center of the frame portion **126**. The frame portion **126** can be made of a metal or plastic, but is not limited to any type of material.

(12) The electronic insect control system **150**, includes a light source **155** (e.g., a fluorescent lamp) used to attract insects. The light source **155** is surrounded by an electrical grid **165**, that can include a wire provided in a spiral configuration around the light source **155**. In some exemplary embodiments, the electrical grid **165** can include one or more wires that can be provided in any configuration and can surround the light source **155** such that the light source **155** is visible through the one or more wires. For example, two or more vertical wires can be provided with a distance or gap between the wires such that the light source **155** is visible through the gap. The gap or distance between adjacent wires can be approximately 2 mm. When in a spiral configuration, the distance between adjacent coils **165** as shown in FIG. **2** can be approximately 2 mm. An insect can be electrocuted by touching the wire(s) of the electrical grid **165** that generates a voltage through the wire(s) high enough (e.g., 2000 volts-4,000 volts) to conduct current through the body of an insect and kill the insect. In some exemplary embodiments, two power modes can be provided for generating a lower voltage (e.g., 2000 volts) and a higher voltage (e.g., 4000 volts).

(13) The electrical grid **165** can provide a high-voltage power supply and can be a transformerless voltage multiplier circuit including diodes and capacitors, generating a voltage high enough (e.g., 2500 volts) to conduct current through the body of an insect, but not high enough to spark across the air gap. Enough electric current is provided to flow through the body of the insect to heat it to a high temperature. The impedance of the power supply and the arrangement of the electrical grid **165** is such that it would not drive a dangerous current through the body of a human.

(14) The frame portion **126** can include several holes **160** around the circumference of the frame portion **126**, which can be long and narrow elliptical holes **160** as shown in FIGS. **1** and **2**. The exemplary embodiments of the present disclosure are not limited to any number, shape or size of the holes **160**. In some exemplary embodiments, the number of holes can range from but is not limited to approximately six to ten holes around the circumference of the frame portion **126**, and can be eight holes. The holes **160** are designed to have a width large enough to allow insects to fly through the holes **160** but small enough so that a human's fingers cannot be inserted through the holes **160**. In some exemplary embodiments, the width W of the holes **160** can range from but is not limited to approximately two to approximately ten millimeters, and preferably approximately four to approximately seven millimeters. The length L of the holes can range from but is not limited to approximately two to approximately four centimeters, and can be approximately three centimeters.

(15) In some exemplary embodiments, a USB port **180** can be provided to recharge the rechargeable batteries provided in, e.g., the lower body portion **124**. Internal circuitry can connect the USB port **180** to the rechargeable batteries inside the flashlight **100**. In some exemplary embodiments, a slider **170** can be provided on the sliding cover **128** shown at the lower body portion **124** in FIG. **3**. The slider **170** can have a flat configuration where a user can press the slider **170** with their finger and slide the sliding cover **128** over the upper body portion **122**, covering the frame portion **126** and electronic insect control system **150**, as shown in FIG. **4**. A user could use the flashlight **100** solely as a flashlight as shown in FIG. **4** without using the electronic insect control system **150**. When desired, the user can slide the sliding cover **128** downward over the lower body portion **124** to use the electronic insect control system **150** of the flashlight **100**. In use,

the user can sway the flashlight **100** with their hand to kill insects while using the flashlight **100**. In some exemplary embodiments, the user can stand the flashlight **100** upright on the tail portion **130**, turning the light **106** off while leaving the electronic insect control system **150** (i.e., light source **155** and electrical grid **165**) on, to use the flashlight **100** solely as a bug zapper.

(16) A switch (not shown) can be provided on the bottom of the tail portion **130**. One switch can be provided to control both the light **106** and the electronic insect control system **150**. For example, circuitry can be provided such that pressing the switch once turns only the light **106** on, pressing the switch again turns only the light source **155** and electrical grid **165** on, pressing the switch again turns the light **106**, light source **155** and electrical grid **165** on, and pressing it again turns the light **106**, light source **155** and electrical grid **165** off. Various configurations of circuitry to operate the light **106** and electronic insect control system **150** are possible and are not limited by the above. In some exemplary embodiments, two switches can be provided on the bottom of the tail portion **130** to operate the light **106** and the electronic insect control system **150** independently. The location of the switches can vary as well, which can be provided on the body portion **120** of the flashlight **100** as well as the tail portion **130** or head portion **110**.

(17) Various other considerations can also be addressed in the exemplary applications described according to the exemplary embodiments of the present disclosure. For example, various materials may be used to construct the elements described in the figures. Various sizes and dimensions of the flashlight **100**, frame portion **126**, electronic insect control system **150** and holes **160** can be provided, and are not limited by the sizes, dimensions, shapes and designs described above and in the figures. In some exemplary embodiments, a dimmer (not shown) can also be provided to dim the light **106** or the light source **155** to vary the brightness of the lights.

(18) Various advantages can be provided based on the exemplary embodiments described above. For example, the flashlight **100** can be used solely as a flashlight, solely as a bug zapper, or both, where the user can use the flashlight **100** while also operating the flashlight as a bug zapper (example, walking at night). The flashlight **100** can be constructed such that it can stand independently on the tail portion **130** or on the head portion **110**, such as when the user wants to use the flashlight **100** solely as a bug zapper.

(19) The words “ranging”, “ranges from”, “ranges between” and other similar notations, are used herein interchangeably and are meant to include the first and second indicated numbers and all the fractional and integral numerals there between. It should be noted that where various embodiments are described by using a given range, the range is given as such merely for convenience and brevity and should not be construed as an inflexible limitation on the scope of the disclosure. Accordingly, the description of a range should be considered to have specifically disclosed all the possible sub-ranges as well as individual numerical values within that range.

(20) The foregoing merely illustrates the principles of the disclosure. Various modifications and alterations to the described embodiments will be apparent to those skilled in the art in view of the teachings herein. It will thus be appreciated that those skilled in the art will be able to devise numerous systems, arrangements, manufacture and methods which, although not explicitly shown or described herein, embody the principles of the disclosure and are thus within the spirit and scope of the disclosure.

Claims

1. A flashlight apparatus, comprising: a first end comprising a first light source; a second end opposite the first end; a body between the first end and the second end, the body having a plurality of vertical elliptical holes around a circumference of the body; a second light source provided within the body; and an electrical grid provided in an interior portion of the body between the vertical elliptical holes and the second light source, the electrical grid provided around the second light source; wherein the electrical grid is configured to generate a voltage.

2. The flashlight apparatus of claim 1, further comprising: one or more batteries configured to power the first light source, second light source and electrical grid.
 3. The flashlight apparatus of claim 2, further comprising: a USB port for charging the one or more batteries.
 4. The flashlight apparatus of claim 1, wherein the first light source comprises a printed circuit board including one or more light emitting diodes.
 5. The flashlight apparatus of claim 1, wherein the second light source comprises a fluorescent lamp.
 6. The flashlight apparatus of claim 1, further comprising: a cover on the body, the cover configured to enclose the body.
 7. The flashlight apparatus of claim 6, wherein the cover is a sliding cover that encloses the plurality of vertical elliptical holes around the circumference of the body in a first position and reveals the plurality of vertical elliptical holes around the circumference of the body in a second position.
 8. The flashlight apparatus of claim 7, further comprising: a slider provided on the sliding cover configured for a user's finger to slide the sliding cover from the first position to the second position.
 9. The flashlight apparatus of claim 1, wherein the body is configured to be held in a user's hand.
 10. The flashlight apparatus of claim 1, wherein the second end is configured to be threadably removed from the body to insert one or more batteries for the flashlight apparatus.
 11. The flashlight apparatus of claim 10, further comprising: a switch at the second end to turn the first light source on and off.
 12. The flashlight apparatus of claim 11, wherein the switch is further configured to turn the electrical grid on and off.
 13. The flashlight apparatus of claim 11, wherein the switch is configured to: (a) turn only the first light source on; (b) turn only the electrical grid on; and (c) turn both the first light source and electrical grid on.
 14. The flashlight apparatus of claim 1, further comprising: a lens cover provided on top of the first light source.
 15. The flashlight apparatus of claim 1, wherein the plurality of vertical elliptical holes around the circumference of the body comprise six to ten vertical elliptical holes.
 16. The flashlight apparatus of claim 15, wherein the plurality of vertical elliptical holes are designed to have a width large enough to allow insects to fly through the plurality of vertical elliptical holes and small enough such that a human's fingers cannot be inserted through the plurality of vertical elliptical holes.
 17. The flashlight apparatus of claim 15, wherein a width of the plurality of vertical elliptical holes are approximately two millimeters to approximately ten millimeters.
 18. The flashlight apparatus of claim 1, wherein the electrical grid comprises a wire provided in a spiral configuration around the second light source.
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